LESSON PLAN

PISTON ENGINE (18)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Titanium and its alloys, copper and its alloys
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Titanium and its alloys, copper and its alloys. Superalloys
5. Specific Objectives	18.1.2.1 Trainee will be able to know the concept of ferrous and non-ferrous metals and their identification etc.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of Titanium and Copper alloys using visual inspection and basic property charts; Safety: Handle samples with care to avoid contamination; Supervision: Instructor monitors identification accuracy.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) Define ferrous and non-ferrous metals.
(b) List two common ferrous metals used in aircraft.
(c) Name one non-ferrous metal known for high strength-to-weight ratio.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and properties of Titanium • Applications of Titanium alloys in aviation • Corrosion resistance of Titanium • Identification characteristics of Titanium	L & Ds	PPT, IB	15 min
(b)	• Properties of Copper and its alloys • Uses of Copper alloys in aircraft systems • Identification of Copper alloys • Comparison of Copper and Titanium properties	L & Ds	PPT, PDF	20 min
(c)	• Overview of Superalloys • Key characteristics of Superalloys • Applications of Superalloys in engines • Distinction between Superalloys and other alloys	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain why Titanium is preferred over steel in aircraft structures.
(b) Differentiate between pure Copper and Brass based on properties.
(c) Identify one critical application of Superalloys in jet engines.

12. Home Assignment and Follow Up	Prepare a brief note on how the knowledge of non-ferrous metals will assist in the upcoming lesson on Heat treatment of Iron and its alloys (LP 08).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Heat treatment of Iron and its alloys. Advantages, procedure and different types of heat treatment	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min